Ultra-Fast Electric Vehicle Charging Systems

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***Abstract*—The rapid adoption of electric vehicles (EV) has caused a significant issue about charging speeds. The development of ultra-fast charging infrastructure has been on the rise delivering high power levels while maintaining safety, efficiency, and battery longevity. This project investigates a 350 kW ultra-fast charging system focusing on thermal management strategies, charging protocol optimization, and techno-economic feasibility. A comprehensive MATLAB/Simulink model is developed to simulate the charging process incorporating a CC-CP-CV (Constant Current-Constant Power-Constant Voltage) charging controller integrated with battery thermal dynamics. Three thermal management approaches are analyzed and compared: liquid cooling, refrigerant-based cooling, and phase change material (PCM) cooling systems. The simulation evaluates thermal performance under extreme charging scenarios, assessing temperature distributions, cooling efficiency, and their impact on charging time and battery degradation. Additionally, an economic analysis examines the capital costs, operational expenses, and revenue potential of deploying 350 kW charging infrastructure. Results demonstrate that effective thermal management is critical for enabling ultra-fast charging rates while preventing thermal runaway and extending battery cycle life. The findings provide insights into optimal cooling strategies and economic viability considerations for next-generation EV charging stations, supporting the transition toward widespread high-power charging infrastructure.**

In this paper, we present a systematic approach to modeling and analyzing ultra-fast charging systems through integrated electrical and thermal simulations. The methodology combines circuit-level charging control with detailed thermal network modeling to capture the complex interactions between charging parameters, heat generation, and cooling system performance under realistic operating conditions.

I. Introduction

The global transition toward electric vehicles (EVs) represents a critical step in reducing greenhouse gas emissions and dependence on fossil fuels. However, widespread EV adoption hinges on the development of charging infrastructure that can match or exceed the convenience of traditional refueling. ultra-fast charging systems, capable of delivering power levels of 350 kW and beyond, have emerged as a promising solution to address range anxiety and reduce charging times to minutes rather than hours [1]. These high-power charging systems present significant technical challenges, particularly in the areas of power electronics design, battery thermal management, and infrastructure economics [2]. One of the most critical challenges in ultra-fast charging is thermal management. High charging currents generate substantial heat within battery cells, which can lead to accelerated degradation, reduced cycle life, and in extreme cases, thermal runaway [3][4]. Effective thermal management strategies are therefore essential to maintain battery temperatures within safe operating limits while enabling rapid energy transfer. Various cooling approaches have been proposed and implemented, including liquid cooling systems, refrigerant-based cooling, and phase change materials (PCMs), each offering distinct advantages in terms of thermal performance, complexity, and cost [5][6][7]. Beyond thermal considerations, the charging protocol itself plays a crucial role in balancing charging speed with battery health. The CC-CP-CV (Constant Current-Constant Power-Constant Voltage) charging strategy has gained attention for its ability to optimize charging time while managing thermal stress [8]. Additionally, understanding battery aging mechanisms and capacity fade under high-power charging conditions is essential for predicting long-term system performance [9][10][11][12]. This project develops a comprehensive simulation framework to analyze ultra-fast EV charging systems, integrating electrical, thermal, and economic models. By comparing different thermal management strategies and evaluating their impact on charging performance and battery longevity, this work aims to provide insights for the design and deployment of next-generation charging infrastructure.

II. Background

The transition to electric mobility has increased dramatically in the past decade;thus, driven by environmental concerns and technological advances in battery systems. However, the availability of charging infrastructure continues to be a key obstacle to the broad adoption of EVs. While conventional Level 2 chargers require 4-8 hours for a full charge, ultra-fast charging systems operating at 350 kW and beyond can reduce charging times to under 15 minutes, approaching the convenience of traditional gasoline refueling [1],[2].

A. EV Charging Standards and Power Levels

EV charging infrastructure is categorized into three primary levels based on power delivery capabilities. Level 1 charging operates at 120V AC, delivering 1.4-1.9 kW, suitable only for overnight residential charging. Level 2 systems utilize 240 V AC with power outputs ranging from 3.3 kW to 19.2 kW, providing practical charging solutions for workplace and

public locations [1]. DC fast charging (DCFC), commonly referred to as Level 3, represents the current state-of-the-art in rapid charging technology.

Contemporary DCFC systems span a wide power range. The CHAdeMO standard supports up to 62.5 kW, while the Combined charging System (CCS) has evolved to enable 350 kW charging. Tesla's Supercharger network currently delivers up to 250 kW, with V4 stations designed for 350 kW operation [2]. The emerging Megawatt charging System (MCS) standard, developed for heavy-duty commercial vehicles, targets power levels exceeding 1 MW.

B. Battery Thermal Dynamics During Ultra-Fast Charging

The primary challenge in ultra-fast charging systems are from the relationship between charging current and heat generation within battery cells. Heat generation in lithium-ion batteries during charging can be expressed as:

$$Q_{\text{gen}} = I^2 R_{\text{internal}} + I \cdot \Delta V_{\text{polarization}} + Q_{\text{reversible}} \tag{1}$$

where I represents charging current, $R_{internal}$ is the cell's internal resistance, and $\Delta V_{polarization}$ accounts for overpotential losses [3]. At 350 kW charging rates for an 800V battery pack, currents can exceed 400A, generating substantial thermal energy that must be dissipated to prevent cell degradation.

Elevated temperatures accelerate multiple degradation mechanisms in lithium-ion batteries. Solid electrolyte interphase (SEI) layer growth increases exponentially with temperature following an Arrhenius relationship:

$$k_{SEI} = A \exp\left(-\frac{E_a}{RT}\right) \tag{2}$$

where E_a is the activation energy, R is the gas constant, and T is absolute temperature [9]. Studies indicate that operation above 50°C can double the rate of capacity fade, while temperatures exceeding 60°C risk thermal runaway.

Temperature distribution within battery packs presents additional challenges. Thermal gradients create non-uniform aging, where cells in hotter zones degrade faster than cooler cells, leading to pack-level imbalances. Research shows that temperature differences exceeding 5°C across a pack can reduce overall cycle life by 20–30% [5].

C. Thermal Management Strategies

1) Liquid Cooling: Represents the most widely adopted thermal management approach for ultra-fast charging applications. These systems circulate coolant (typically water-glycol mixtures) through channels integrated into battery pack structures. The heat transfer relationship is governed by:

$$Q_{transfer} = \dot{m} c_p (T_{out} - T_{in}) = hA\Delta T_{log-mean} \tag{3}$$

where $\dot{m}$ is mass flow rate, c_p is specific heat capacity, h is the convective heat transfer coefficient, and A is the heat transfer surface area [6].

Liquid cooling systems can achieve heat transfer coefficients of 500–1500 W/m^{2}K, enabling effective thermal control even at high charging rates. Cold plates with optimized channel geometries can maintain cell temperatures below 45°C during 350 kW charging, though this requires pumping power of 200–500 W.

2) Refrigerant-Based Cooling: Direct refrigerant cooling offers superior thermal performance by exploiting phase-change heat transfer. Two-phase cooling systems achieve heat transfer coefficients exceeding 3000 W/m^{2}K through evaporative cooling. The cooling capacity is expressed as:

$$Q_{cooling} = \dot{m}_{ref} h_{fg} \tag{4}$$

where $\dot{m}_{ref}$ is refrigerant mass flow rate and h_{fg} is the latent heat of vaporization [11].

Refrigerant systems can maintain more uniform temperature distributions across large battery packs and respond more rapidly to transient thermal loads. However, they require complex refrigeration circuits, increasing system cost and parasitic power consumption.

3) Phase Change Material (PCM) Cooling: PCM-based thermal management utilizes materials that absorb large quantities of heat during melting transitions. Common PCMs for battery applications include paraffin waxes with melting points between 40–50°C. The energy storage capacity is:

$$Q_{stored} = m_{PCM}(c_p \Delta T + h_{fusion}) \tag{5}$$

where m_{PCM} is PCM mass, and h_{fusion} is the latent heat of fusion [5].

PCMs offer passive thermal buffering without pumps or compressors, but their effectiveness diminishes after the material fully melts. Hybrid systems combining PCM with active cooling show promise for managing intermittent ultra-fast charging events while minimizing active cooling requirements during normal operation.

D. Charging Protocols and Control Strategies

The CC-CP-CV (Constant Current-Constant Power-Constant Voltage) charging protocol has emerged as the preferred strategy for ultra-fast charging. This three-stage approach optimizes charging speed while managing thermal and electrical stress [8].

In the constant current (CC) phase, the charger delivers maximum current until either the battery voltage reaches a threshold (typically 80–90% of maximum voltage) or thermal limits are approached. The charging current is constrained by:

$$I_{max} = \min\left(\frac{P_{charger}}{V_{battery}}, I_{thermal-limit}, I_{C-rate}\right) \tag{6}$$

The constant power (CP) phase maintains maximum power delivery as voltage increases, allowing current to decrease naturally. This phase bridges CC and CV operation:

$$P = V_{battery} \cdot I_{charging} = \text{constant} \tag{7}$$

Finally, the constant voltage (CV) phase holds voltage at the cell maximum while current tapers exponentially, ensuring complete charge while preventing overvoltage damage.

Advanced charging algorithms incorporate real-time thermal feedback, adjusting charging current based on cell temperature measurements and state-of-charge estimations. Model predictive control (MPC) approaches optimize the charging trajectory by solving:

$$\min_{I(t)} \int_0^{t_f} \left(w_1 \cdot t_f + w_2 \cdot \int_0^{t_f} (T - T_{ref})^2 \, dt \right) \tag{8}$$

subject to thermal and electrical constraints, where w_1 and w_2 are weighting factors for charging time and thermal deviation [8].

E. Infrastructure and Economic Considerations

Deploying 350 kW charging infrastructure requires significant electrical infrastructure upgrades. A single ultra-fast Charger draws current comparable to a small commercial building, necessitating dedicated high-voltage connections and transformers. Installation costs typically range from $150,000 to $300,000 per charging stall, including electrical infrastructure and civil works [2].

Grid integration presents additional challenges. Multiple simultaneous charging events can create demand peaks exceeding megawatts. Energy storage systems (ESS) co-located with charging stations can buffer these peaks, reducing demand charges and enabling installation in grid-constrained locations. The economic optimization balances ESS capacity against utility rate structures:

$$\text{Total Cost} = C_{energy} + C_{demand} + C_{ESS} - R_{utilization} \tag{9}$$

where C_{energy} represents energy charges, C_{demand} accounts for demand charges, C_{ESS} is the annualized energy storage cost, and $R_{utilization}$ reflects revenue from charging services.

Revenue models for ultra-fast charging Stations depend on utilization rates, pricing strategies, and operational costs [8].

III. Methodology

A. Overview of the Fast-Charging Approach

The fast-charging strategy implemented in this work is based on the widely adopted constant-current/constant-voltage (CC–CV) method, which provides an effective balance between charging speed, efficiency, and battery safety. The battery pack is charged from a three-phase AC grid through a controlled AC–DC active rectifier followed by a high-power DC–DC Buck-Boost Converter. This two-stage approach enables independent regulation of the charging current and battery voltage. The maximum charging power is limited to 350 kW, corresponding to ultra-fast charging conditions.

The overall fast-charging system consists of:

- A three-phase active rectifier to control grid-side currents and maintain unity power factor.
- A DC–DC Buck-Boost Converter that regulates the battery charging current and voltage.

The inner current control loop operates at a bandwidth significantly lower than the switching frequency to ensure stability, whereas the outer voltage and SOC-based control loops operate more slowly. Proportional–integral (PI) controllers are employed for both inner and outer loops. The gains are selected based on converter inductance, resistance, and desired crossover frequencies.

B. System Architecture

The fast-charging system utilizes a two-stage power conversion architecture composed of a three-phase active AC–DC rectifier and a DC–DC Buck-Boost Converter. This structure provides independent control of grid-side power quality and battery-side charging dynamics. The rectifier stage maintains controlled energy flow to and from the utility grid, while the DC–DC stage regulates battery charging current and voltage to follow the CC–CV charging strategy.

C. Rectifier Control Strategy

The AC–DC rectifier employs a voltage-oriented control (VOC) scheme in the synchronous dq reference frame. The primary control objectives are:

1) Regulation of the DC-link voltage to a constant reference of 800 V.
2) Alignment of grid-side currents with the grid voltage to ensure unity power factor.

An outer DC-link voltage loop generates the reference $i_d^\star$ for the d-axis current, while an inner current loop regulates the d- and q-axis currents using PI controllers. The q-axis current reference is set to zero to eliminate reactive power flow. This hierarchical control structure ensures fast dynamic response and a stable DC-link voltage under varying load conditions.

D. Buck-Boost DC–DC Converter

The DC–DC converter interfaces the regulated DC-link voltage (800 V) with the battery pack, whose voltage varies widely from 350–450 V depending on SOC. The Buck-Boost topology is selected due to its ability to step down voltage, satisfying battery voltage requirements throughout the entire charging cycle.

1) charging Control Implementation: The converter operates in two charging modes:

a) Constant-Current (CC) Mode: At low to medium SOC, the converter regulates battery current to a constant reference, determined by the charging power limit. This enables rapid charging while staying within battery safety limits.

TABLE I: Key Electrical Parameters Used in the Model

Parameter	Value	Notes
Battery energy	100 kWh	Pack nominal energy
Pack nominal voltage	350–450 V	Used for CC/CV logic
Constant current in CC	437.5 A	Used for CC mode (350 kW)
Constant voltage in CV	450 V	Used for CV mode
Battery mass m	500 kg	From project image

TABLE II: Key Thermal Parameters Used in the Model

Parameter	Value	Notes
Battery specific heat c_p	1000 J/kg·°C	$mc_p = 5\times10^5$ J/°C
Battery thermal capacitance $C_{\text{th},b}$	5×10^5 J/°C	Lumped pack thermal mass
Surface area A	3 m^2	Cold-plate contact area (estimate)
Heat transfer coefficient h	≈ 50 W/m^2°C	$hA \approx 150$ W/°C
Coolant inlet temperature	20°C	Glycol–water mixture set-point
Ambient temperature	25°C	Environmental temperature
Coolant lumped thermal capacitance $C_{\text{th},c}$	500–5000 J/°C	Depends on coolant volume; default 500 J/°C
Internal resistance R_{int}	0.002–0.01 Ω	Typical lithium-ion range

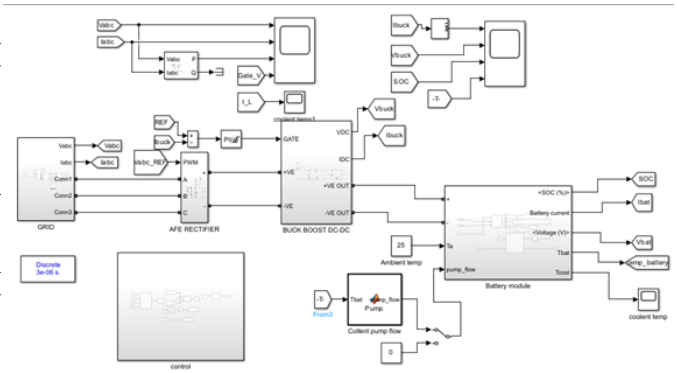


Fig. 1: Overall Simulink model of the ultra-fast battery charging system integrating the three-phase grid, active AC–DC rectifier, DC–DC Buck-Boost Converter, battery model, and thermal management subsystem.

b) Constant-Voltage (CV) Mode: When battery SOC exceeds 80%, the controller transitions to voltage regulation mode. The battery voltage is held at its maximum allowable level (450 V), causing the charging current to naturally taper as SOC increases. This reduces thermal stress and prevents overvoltage.

SOC-based logic ensures a smooth and stable transition between CC and CV modes.

E. Key Parameters Used in the Model

F. Battery Heat Generation Model

The dominant source of battery heat during fast charging is Joule heating caused by internal resistance. The heat generation rate is modeled as:

$$Q_{\text{gen}} = I_{\text{bat}}^2 R_{\text{int}}, \tag{10}$$

where I_{bat} is the battery charging current and R_{int} is the battery internal resistance. This formulation is widely used for system-level thermal analysis under high-power charging conditions.

G. Lumped Thermal Model of the Battery

The battery pack is modeled as a lumped thermal mass with uniform temperature. The thermal dynamics are given by:

$$C_{\text{th},b}\frac{dT_{\text{bat}}}{dt} = Q_{\text{gen}} - Q_{\text{cool}} - Q_{\text{amb}}, \tag{11}$$

where:

$$C_{\text{th},b} = mc_p,$$

T_{bat} is the battery temperature, Q_{cool} is heat removed by the coolant, and Q_{amb} is heat lost to ambient air.

H. Liquid Cooling Model

A liquid-cooled cold plate is used to manage thermal behavior during charging. Heat transfer between the battery and coolant follows Newtonâs law of convection:

$$Q_{\text{cool}} = hA\left(T_{\text{bat}} - T_{\text{cool}}\right), \tag{12}$$

where T_{cool} is the coolant temperature. Coolant inlet temperature is regulated at 20°C to ensure consistent thermal performance.

The coolant dynamics are represented using a lumped thermal capacitance:

$$C_{\text{th},c}\frac{dT_{\text{cool}}}{dt} = Q_{\text{cool}}. \tag{13}$$

I. Simulink Model Description

1) Overall Model Structure: The complete fast-charging system is implemented in MATLAB/Simulink with modular subsystems representing the grid interface, AC–DC rectifier, DC–DC converter, battery electrical model, and battery thermal model. This integrated representation enables simultaneous evaluation of electrical and thermal behavior under realistic fast-charging conditions.

This section describes the modeling and control design procedure adopted for the DC fast charging System shown in Fig. 1. The system consists of a three-phase grid, an active front-end (AFE) rectifier with LCL filter, a DC-link capacitor, a DC–DC Buck-Boost Converter, and a lithium-ion battery module with thermal management.

J. Three-Phase Grid Model

The three-phase grid is modeled as an ideal balanced voltage source operating at 400 V (line-to-line RMS) and 50 Hz. Grid impedance is included to represent practical connection conditions and to emulate the behavior of a realistic distribution feeder. This subsystem provides the AC input to the charging station and serves as the reference for synchronizing the active rectifier control.

K. Three-Phase Active Rectifier

The active rectifier converts the three-phase AC grid voltage into a regulated DC-link voltage. It is implemented using a controlled IGBT-based converter with an LC filter to reduce current harmonics. The rectifier control maintains the DC-link voltage at 800 V while ensuring unity power factor operation by aligning grid currents with grid voltages.

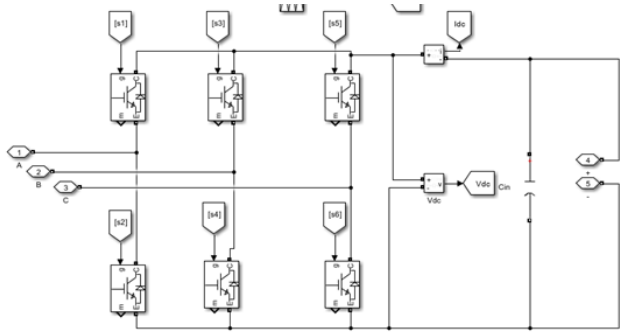


Fig. 2: Three-phase grid and active rectifier subsystem used to convert AC grid power into a regulated DC-link voltage under unity power factor operation.

L. DC-DC Buck-Boost Converter

The DC Buck-Boost converter interfaces the DC link with the battery pack and enables regulation of the charging current and voltage over a wide battery voltage range. This stage is responsible for implementing the CC-CV charging strategy by controlling the inductor current. The converter ensures smooth transition between constant-current and constant-voltage modes based on battery state of charge.

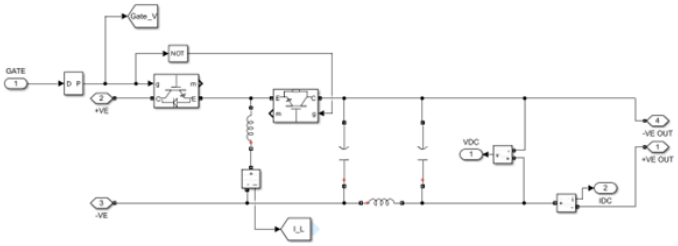


Fig. 3: DC–DC Buck-Boost Converter implementing constant-current–constant-voltage (CC–CV) control for regulated battery charging over a wide voltage range.

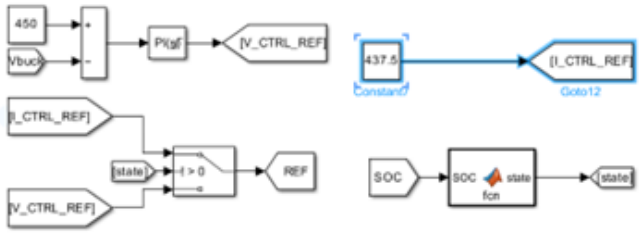


Fig. 4: CC–CV control methodology for the DC Buck-Boost Converter.

M. Battery Model

The battery is modeled using an equivalent electrical circuit combined with a lumped thermal model. The electrical model captures voltage and current behavior during fast charging, while the thermal model computes battery temperature based on internal heat generation and cooling conditions. The battery subsystem includes inputs for ambient temperature and cooling interaction, enabling evaluation of both cooled and uncooled charging scenarios.

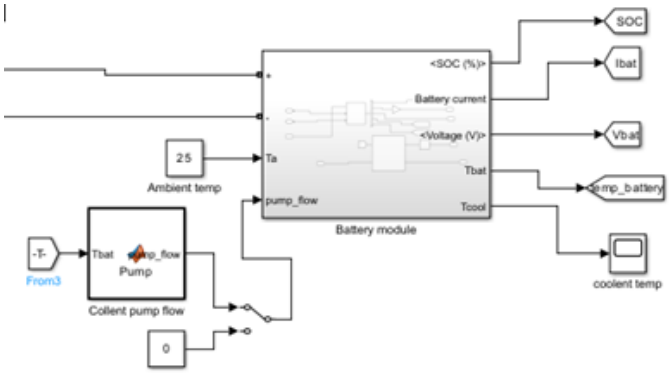


Fig. 5: Battery electrical and thermal model with electro-thermal coupling and liquid cooling interface for temperature regulation during fast charging.

N. System Specification

The charger is rated at

$$P = 350 \text{ kW}, \tag{14}$$

and is connected to a three-phase grid with line-to-line RMS voltage and frequency

$$V_{\text{LL}} = 400 \text{ V}, \qquad f_{\text{g}} = 50 \text{ Hz}. \tag{15}$$

O. Simulation Setup

The complete system is implemented in MATLAB/Simulink using discrete-time power electronics models (with a Powergui block configured for a sampling time of 30 μs) and continuous-time control blocks. Unless otherwise stated, the simulation horizon is set to

$$T_{\text{sim}} = 5 \text{ s}, \tag{16}$$

which is sufficient to observe the transient response of the DC-link voltage, grid currents, and battery charging variables under step changes in power and SOC.

IV. Results and Discussion

This section presents the electrical and thermal responses of the fast-charging system under 350 kW charging conditions. The results demonstrate proper operation of the AC–DC rectifier, DC–DC Buck-Boost Converter, battery charging controller, and the thermal management system.

A. Inductor Current of the Buck-Boost Converter

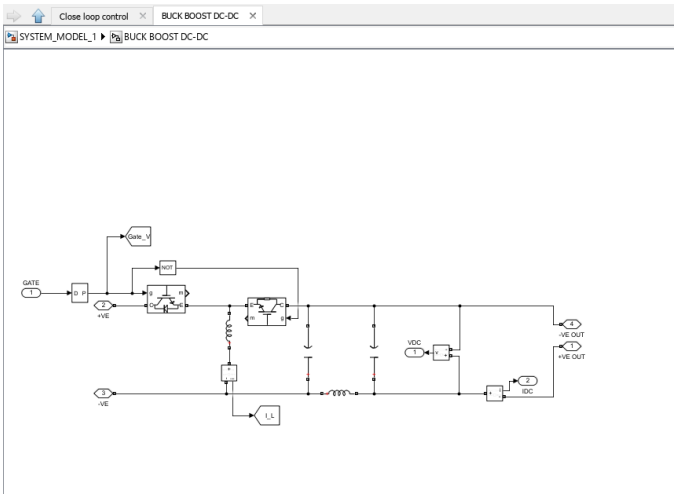


Fig. 6: Inductor Current Measurement

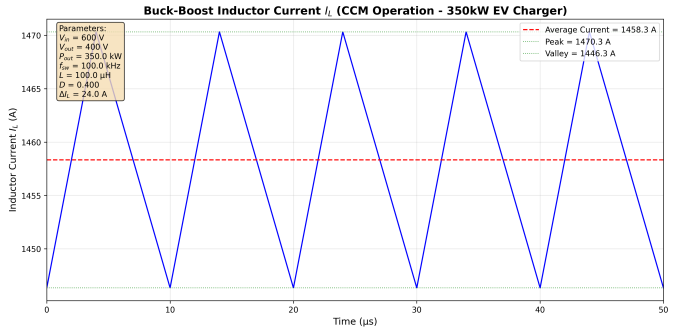


Fig. 7: Inductor Current Plot with respect to time.

Fig.7 shows the zoomed inductor current I_L of the DC–DC Buck-Boost converter during steady-state operation. The waveform exhibits the expected triangular shape, corresponding to the switching dynamics imposed by the converter operating at 100 kHz. The peak current reaches approximately 1.5 kA, while the minimum current approaches zero, confirming that the converter operates near the boundary of continuous conduction mode under high-power charging. The ripple magnitude is consistent with the inductor design specifications and the chosen switching frequency, validating the sizing of the converter inductor.

B. DC-Link Voltage Regulation

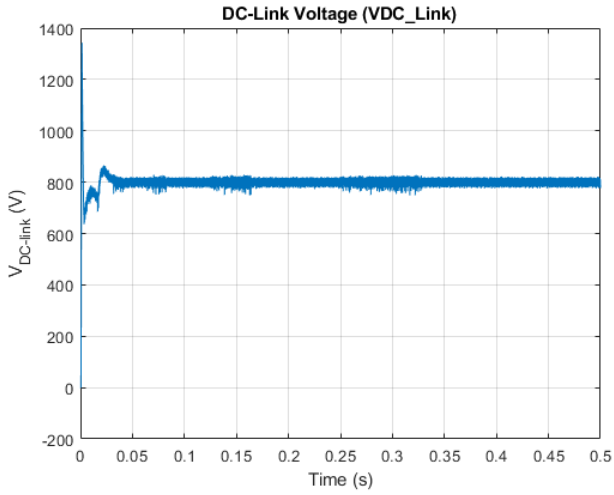


Fig. 8: DC–Link Voltage Plot with respect to time.

The regulated DC-link voltage, shown in Fig.8, remains tightly controlled around 800 V after initial transients. A brief overshoot occurs during the first 20–30 ms as the active rectifier establishes synchronous operation with the grid. Once the rectifier and outer voltage loop settle, the DC-link voltage maintains steady-state operation with minimal ripple. This confirms effective performance of the voltage-oriented control (VOC) strategy and demonstrates proper decoupling between the rectifier and DC–DC conversion stages.

C. Grid Voltage and Current Waveforms

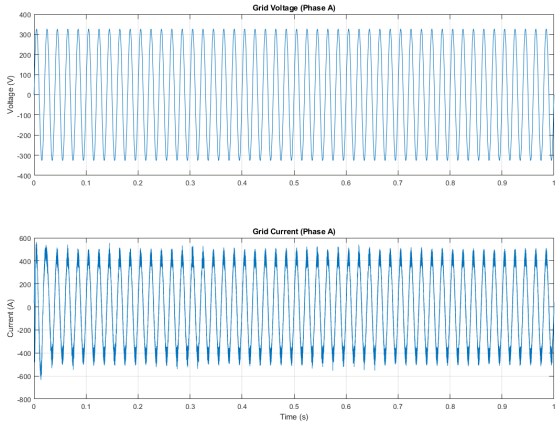


Fig. 9: Phase A Grid Voltage and Current Plots.

Fig.9 presents the measured phase-*A* grid voltage and current. The grid voltage is a clean sinusoid at 50 Hz with amplitude ≈ 325 V (phase-to-neutral). The grid current waveform is also sinusoidal and nearly in phase with the voltage, validating unity power factor operation of the active rectifier. Minor high-frequency content in the current waveform is caused

by switching ripple from the PWM-controlled rectifier. The results confirm that the LCL filter and current control loops effectively attenuate harmonics while ensuring high-quality grid interaction.

D. Coolant Temperature Response

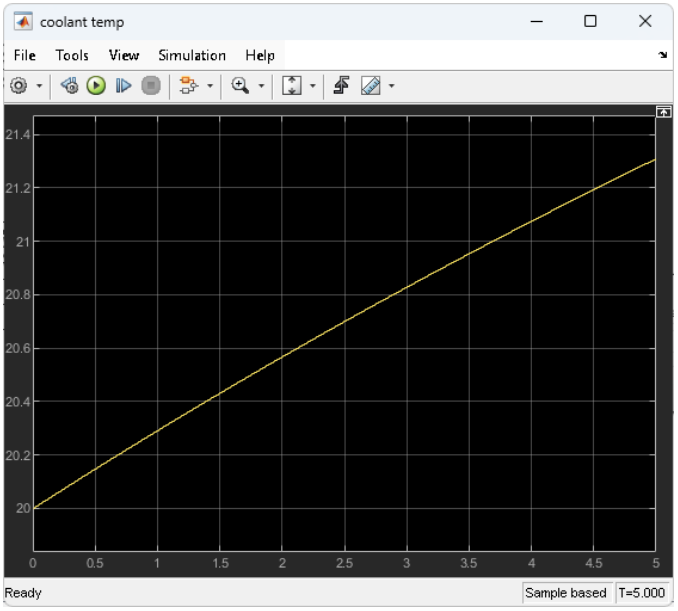


Fig. 10: Coolant Temperature Plot.

The coolant temperature evolution during charging is shown in Fig.10. Over the 5 s simulation interval, the coolant temperature rises gradually from 20°C to approximately 21.3°C. This mild temperature increase reflects the relatively low thermal mass of the coolant but also demonstrates that the cooling loop is functioning properly. Since only a short-duration charging interval is modeled, the coolant system remains within safe operating limits. Longer charging durations would produce a more pronounced rise in coolant temperature, which the thermal management system is designed to regulate.

E. Battery charging Profiles

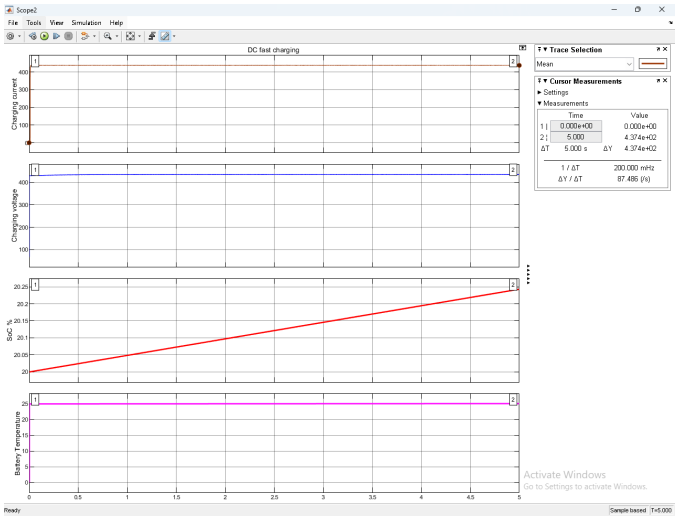


Fig. 11: Scope 2: Measures charging Current, charging Voltage, State of Charge, and Battery Temperature

Battery charging characteristics are shown in Fig.11. The charging current remains at approximately 437A during the constant-current (CC) charging stage. The battery terminal voltage increases only slightly over the 5s window, as expected during early CC-mode operation when SOC is relatively low.

The state of charge increases from 20% to 20.25%, demonstrating correct integration of the battery energy model. The battery temperature remains effectively constant at 25°C due to the short simulation duration and the activation of the coolant loop. These results confirm that the CC–CV charging strategy and thermal subsystem behave consistently with ultra-fast charging requirements.

Overall, the combined electrical and thermal results verify robust operation of the fast-charging architecture, including stable DC-link regulation, unity power factor grid interaction, correct inductor behavior within the Buck-Boost Converter, and effective thermal management. The system operates within the expected design limits for both electrical waveforms and temperature evolution.

V. Conclusion

The combination of a three-phase active rectifier and a Buck-Boost DC-DC Converter provides an effective and flexible power conversion solution for ultra-fast battery charging. The rectifier ensures grid compliance and stable DC-link operation, while the DC-DC converter enables precise implementation of CC charging with 437.5A and CV with 450V charging. Together, these stages form the electrical foundation upon which electro-thermal performance and cooling strategies are evaluated in this study. The Simulink model provides a comprehensive and flexible platform for analyzing high-power fast charging with integrated thermal management. The modular structure allows individual subsystems to be

studied independently while preserving realistic system-level interactions.

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APPENDIX A
NOMENCLATURE AND MODEL SYMBOLS

This appendix summarizes the main symbols used in the electrical and thermal models of the ultra-fast charging system.

- P : Charger rated power (W).
- V_{LL} : Three-phase grid line-to-line RMS voltage (V).
- f_{g} : Grid frequency (Hz).
- V_{dc} : DC-link voltage (V).
- V_{bat} : Battery terminal voltage (V).
- I_{bat} : Battery charging current (A).
- I_L : Inductor current of the Buck-Boost converter (A).
- R_{int} : Battery internal resistance (Ω).
- m : Battery pack mass (kg).
- c_p : Battery specific heat (J/kg·°C).
- $C_{\mathrm{th},b} = mc_p$: Battery thermal capacitance (J/°C).
- $C_{\mathrm{th},c}$: Coolant lumped thermal capacitance (J/°C).
- T_{bat} : Battery temperature (°C).
- T_{cool} : Coolant temperature (°C).
- T_{amb} : Ambient temperature (°C).
- A : Effective contact/heat transfer area between battery and cold plate (m^2).
- h : Convective heat transfer coefficient ($\mathrm{W/m^2{\circ}C}$).
- Q_{gen} : Heat generated inside the battery (W).
- Q_{cool} : Heat removed by liquid cooling (W).
- Q_{amb} : Heat lost to ambient air (W).
- SOC : Battery state-of-charge (fraction or %).

The principal thermal and electrical relationships used in the model are

$$Q_{\mathrm{gen}} = I_{\mathrm{bat}}^2 R_{\mathrm{int}}, \tag{17}$$

$$C_{\mathrm{th},b}\frac{dT_{\mathrm{bat}}}{dt} = Q_{\mathrm{gen}} - Q_{\mathrm{cool}} - Q_{\mathrm{amb}}, \tag{18}$$

$$Q_{\mathrm{cool}} = hA\,(T_{\mathrm{bat}} - T_{\mathrm{cool}}), \tag{19}$$

$$C_{\mathrm{th},c}\frac{dT_{\mathrm{cool}}}{dt} = Q_{\mathrm{cool}}. \tag{20}$$

APPENDIX B
KEY ELECTRICAL AND THERMAL PARAMETERS

For convenience, the main model parameters used in the simulations are repeated here.

TABLE III: Electrical Parameters Used in the Simulink Model

Parameter	Value	Notes
Charger rated power P	350 kW	DC fast charger
Grid voltage V_{LL}	400 V	3–phase, line-to-line RMS
Grid frequency f_{g}	50 Hz	Utility grid
DC-link nominal voltage V_{dc}	800 V	Rectifier output
Battery voltage range	350–450 V	SOC-dependent
CC charging current	437.5 A	$\approx$ 350 kW at 800 V
CV charging voltage	450 V	Max battery voltage
Switching frequency f_{sw}	100 kHz	Buck-Boost converter

TABLE IV: Thermal Parameters Used in the Model

Parameter	Value	Notes
Battery energy	100 kWh	Pack nominal energy
Battery mass m	500 kg	From project data
Specific heat c_p	1000 J/kg·°C	Li-ion cells (typical)
Battery thermal capacitance $C_{\mathrm{th},b}$	5×10^5 J/°C	mc_p
Surface area A	3 m^2	Cold-plate contact area
Heat transfer coefficient h	$\approx$ 50 $\mathrm{W/m^2{\circ}C}$	Liquid cold plate
Coolant inlet temperature	20°C	Setpoint
Ambient temperature T_{amb}	25°C	Room air
Coolant thermal capacitance $C_{\mathrm{th},c}$	500–5000 J/°C	Depends on volume
Internal resistance R_{int}	0.002–0.01 Ω	Li-ion range